

HOHO Hotel

DATABASE MANAGEMENT SYSTEM

Group1

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INTRODUCTION

Project Context

- Rapidly expanding hotel chain
- Large complexes with rooms, facilities, events, services, and billing
- Staff need current, complete information

Core Functions Needed

- Rooms have flexible uses and changing availability
- Guests, hosts, organizations, and billing parties can overlap
- Charges must be tracked in real time

Project Goal

- Design a relational database to organize hotel operations
- Model basic compositions, rooms, reservations, events, guests, services, and bills
- Build queries and a web interface to turn records into useful summaries

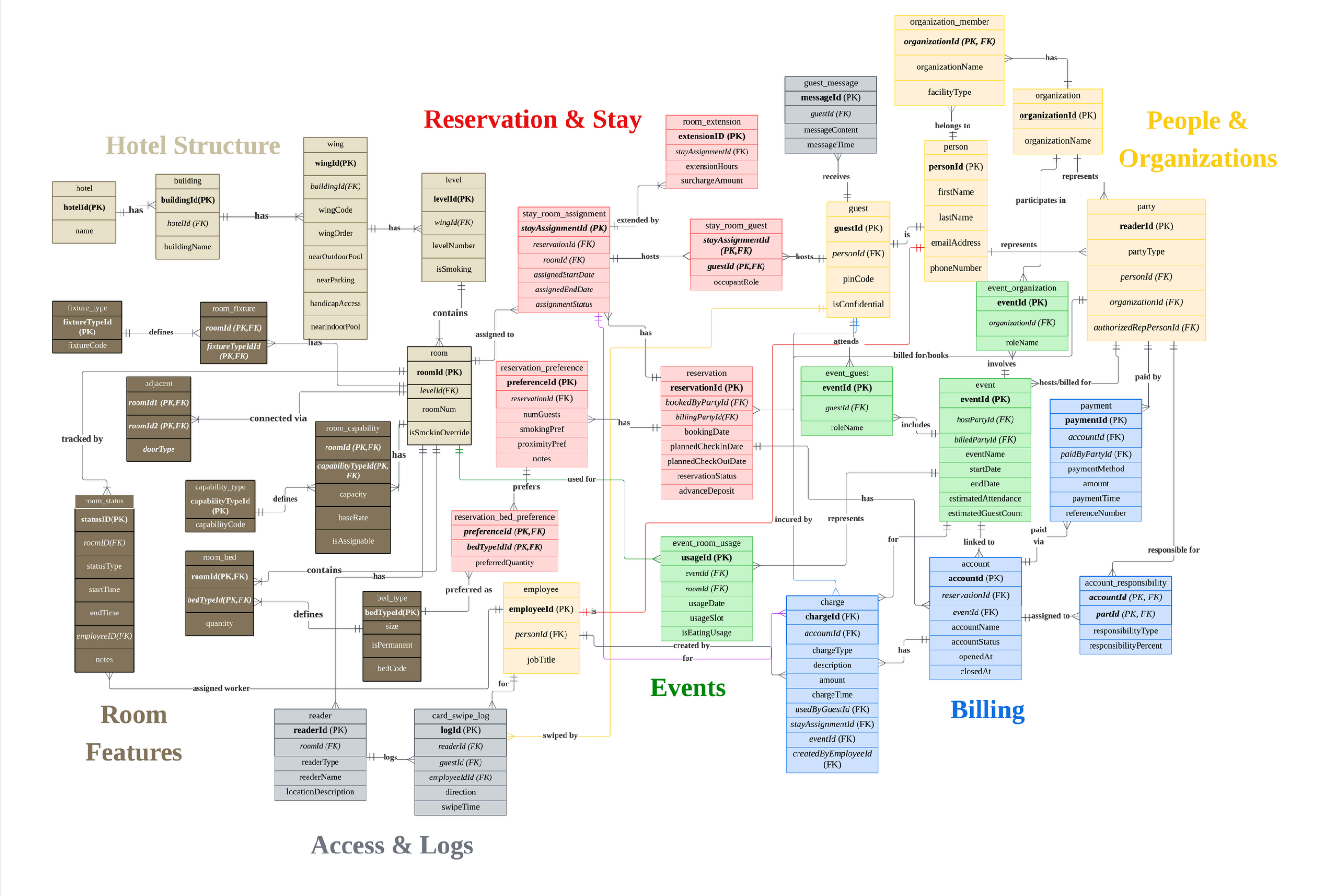


Credit to Google NanoBanana 2



ERD

7 sections36 tables



ERD + SCHEMA DESIGN

7 sections 36 tables

Hotel Structure

- hotel
- building
 - nearIndoorPool INTEGER NOT NULL DEFAULT 0 CHECK (nearIndoorPool IN (0, 1)),
 - nearOutdoorPool INTEGER NOT NULL DEFAULT 0 CHECK (nearOutdoorPool IN (0, 1)),
 - nearParking INTEGER NOT NULL DEFAULT 0 CHECK (nearParking IN (0, 1)),
 - handicapAccess INTEGER NOT NULL DEFAULT 0 CHECK (handicapAccess IN (0, 1)),
- wing
- level
 - isSmoking INTEGER NOT NULL CHECK (isSmoking IN (0, 1)),
- room
 - isSmokingOverride INTEGER DEFAULT NULL CHECK (isSmokingOverride IN (0, 1)),

```
CREATE TABLE room_status (  
  statusId INTEGER PRIMARY KEY,  
  roomId INTEGER NOT NULL,  
  statusType TEXT NOT NULL CHECK (  
    statusType IN (  
      'AVAILABLE',  
      'OCCUPIED',  
      'DIRTY',  
      'READY',  
      'RENOVATION',  
      'RECONSTRUCTION',  
      'OUT_OF_SERVICE'  
    )  
  ),  
  startTime TIMESTAMP NOT NULL,  
  endTime TIMESTAMP,  
  employeeId INTEGER,  
  notes TEXT,  
  FOREIGN KEY (roomId) REFERENCES room(roomId),  
  CHECK (endTime IS NULL OR endTime >= startTime)  
);
```

Room Features

- capability_type
- room_capability
- fixture_type
- room_fixture
- bed_type
- room_bed
- adjacent
- room_status

People & Organizations

- person
- guest
- employee
- organization
- organization_member
- party

```
CREATE TABLE party (  
  partyId INTEGER PRIMARY KEY,  
  partyType TEXT NOT NULL CHECK (partyType IN ('PERSON', 'ORGANIZATION')),  
  personId INTEGER,  
  organizationId INTEGER,  
  authorizedRepPersonId INTEGER,  
  FOREIGN KEY (personId) REFERENCES person(personId),  
  FOREIGN KEY (organizationId) REFERENCES organization(organizationId),  
  FOREIGN KEY (authorizedRepPersonId) REFERENCES person(personId),  
  CHECK (  
    (partyType = 'PERSON' AND personId IS NOT NULL AND organizationId IS NULL)  
    OR  
    (partyType = 'ORGANIZATION' AND organizationId IS NOT NULL AND personId IS NULL)  
  )  
);
```

Reservation & Stay

- reservation
- reservation_preference
- reservation_bed_preference
- stay_room_assignment
- stay_room_guest
- room_extension

Billing

- account
- account_responsibility
- charge
- payment

Events

- event
- event_guest
- event_organization
- event_room_usage

Access & Logs

- reader
- card_swipe_log
- guest_message

```
CREATE TABLE event_room_usage (  
  usageId INTEGER PRIMARY KEY,  
  eventId INTEGER NOT NULL,  
  roomId INTEGER NOT NULL,  
  usageDate DATE NOT NULL,  
  usageSlot TEXT NOT NULL CHECK (  
    usageSlot IN ('BREAKFAST', 'MORNING', 'LUNCH', 'AFTERNOON', 'SUPPER', 'EVENING', 'NIGHT')  
  ),  
  isEatingUsage INTEGER NOT NULL CHECK (isEatingUsage IN (0, 1)),  
  FOREIGN KEY (eventId) REFERENCES event(eventId),  
  FOREIGN KEY (roomId) REFERENCES room(roomId),  
  UNIQUE (roomId, usageDate, usageSlot)  
);
```

```
CREATE TABLE account_responsibility (  
  accountId INTEGER NOT NULL,  
  partyId INTEGER NOT NULL,  
  responsibilityType TEXT NOT NULL CHECK (  
    responsibilityType IN ('FULL', 'SPLIT', 'BACKUP')  
  ),  
  responsibilityPercent DECIMAL(5, 2),  
  PRIMARY KEY (accountId, partyId),  
  FOREIGN KEY (accountId) REFERENCES account(accountId),  
  FOREIGN KEY (partyId) REFERENCES party(partyId),  
  CHECK (  
    responsibilityPercent IS NULL  
    OR (responsibilityPercent >= 0 AND responsibilityPercent <= 100)  
  )  
);
```

```
CREATE TABLE card_swipe_log (  
  logId INTEGER PRIMARY KEY,  
  readerId INTEGER NOT NULL,  
  guestId INTEGER,  
  employeeId INTEGER,  
  direction TEXT NOT NULL CHECK (direction IN ('IN', 'OUT')),  
  swipeTime TIMESTAMP NOT NULL,  
  FOREIGN KEY (readerId) REFERENCES reader(readerId),  
  FOREIGN KEY (guestId) REFERENCES guest(guestId),  
  FOREIGN KEY (employeeId) REFERENCES employee(employeeId),  
  CHECK (  
    (guestId IS NOT NULL AND employeeId IS NULL)  
    OR  
    (guestId IS NULL AND employeeId IS NOT NULL)  
  )  
);
```



Q U E R I E S

Operations

- Current guests + room + affiliations
- Active events + assigned rooms

Room Management

- Dynamic availability (booking + maintenance)
- Room status summary & trends

Billing

- Account balances (charges – payments)
- Revenue breakdown by type

Guest Tracking

- Last known location (card swipes)



Web Demo



Thank you!

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Hotel

